Recall Tensor Product (张量积)

Question Can you define $V \boxtimes W$ s.t. $\dim V \boxtimes W = \dim V \cdot \dim W$ $\left\{\begin{array}{l} \text{product = sum} \\ \text{Quotient = substract} \end{array}\right)$

$\boxtimes$ $\odot$ not so good!

1950, Cartier Existence + Uniqueness

Def Let $V, W / \mathbb{F}$ $\quad t: V \times W \longrightarrow T$ bilinear. If $\forall X / \mathbb{F}$,

$\forall b: V \times W \longrightarrow X$ bilinear $\exists! \underset{\text{universal map}}{u}: T \longrightarrow X$ linear s.t. $b = ut$

Then (T, t) is a tensor product

Thm Tensor Product is unique up to isomorphism

Pf. Let (T, t) and (T', t') be two tensor products of V and W

Then
$$\begin{tikzcd} V \times W \arrow[r, "t"] \arrow[dr, "t'"'] & T \arrow[d, shift right, "\exists! u"'] \\ & T' \arrow[u, shift right, "\exists! u'"'] \end{tikzcd}$$

$t' = ut$ (Denoted as $V \otimes W$)

$t = u't' = u'ut$

$$\begin{tikzcd} V \times W \arrow[r, "t"] \arrow[dr, "t"'] & T \arrow[d, "u'u"] \\ & T \end{tikzcd}$$

However
$$\begin{tikzcd} V \times W \arrow[r, "t"] \arrow[dr, "t"'] & T \arrow[d, "\exists!\, u_T = 1"] \\ & T \end{tikzcd}$$

According to the uniqueness of the universal map.

we have $u'u = I_T \Rightarrow$ u' is surj (epic), u is inj (monic)

若在矩阵中，u' 行满秩，u 列满秩

Similarly, u' is inj, u is surj.

So both u and u' are isomorphisms

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Ex 1

$$\begin{tikzcd}
\mathbb{F}^m \otimes \mathbb{F}^n \arrow[d, "\exists! u"'] & \mathbb{F}^m \times \mathbb{F}^n \arrow[l, "t"'] \arrow[dl, "\forall b"] \\
\operatorname{span}\{(e_i, f_j)\} = \bigoplus_{j=1}^{n}\bigoplus_{i=1}^{m} \mathbb{F}(e_i, f_j) &
\end{tikzcd}$$

$t(x,y)$ $\qquad (x,y) = (\sum_{i=1}^{m} x_i e_i, \sum_{j=1}^{n} y_j f_j)$

(e_i, e_j)

$b = ut$ $\qquad b(x,y) = (x,y)$

$0 = u(\tau) = u(\Sigma t(x,y))$

$= \Sigma b(x,y) = 0 = \Sigma (x,y)$

$\Rightarrow \tau = 0$

① $m = n = 1$

$$\begin{tikzcd}
\underset{(x,y)}{\mathbb{F}^1 \times \mathbb{F}^1} \arrow[r, "t"] \arrow[dr, "\forall b"'] & \mathbb{F}^1 \otimes \mathbb{F}^1 \ni \tau \arrow[d, "\exists! u"] \\
& \operatorname{span}\{(e_1, f_1)\} = \mathbb{F}(e_1, f_1)
\end{tikzcd}$$

$1 = e_1$, $f_1 = \lambda$

$t(x,y)$

span Im t $= \Sigma t(x_i, y_i)$

$u\tau = u(\Sigma t(x_i, y_i))$

$= \Sigma u t(x_i, y_i)$

$= \Sigma b(x_i, y_i)$

$b(x,y) = b(x \cdot 1, \frac{y}{\lambda} \cdot \lambda) = \frac{xy}{\lambda} b(1, \lambda)$

- Question ① What about t?

$\ker t = \{(x,y) \in \mathbb{F}^1 \times \mathbb{F}^1 \mid t(x,y) = 0\}$

$\operatorname{Im} t = \{\tau \in \mathbb{F}^1 \otimes \mathbb{F}^1 \mid \tau = t(x,y),\ \exists (x,y) \in \mathbb{F}^1 \times \mathbb{F}^1\}$

$t(x_1, y_1) + t(x_2, y_2) = t(\bar{x}, \bar{y})$ $\qquad$ t is not surjective

$\lambda t(x,y) = t(\lambda x, y) = t(x, \lambda y)$

Observation $\operatorname{Im} t \subsetneq T \Rightarrow \operatorname{Span} \operatorname{Im} t = T$

Ex 2

$$\begin{tikzcd}
V \otimes W \arrow[r, "u"] & F(V \times W)/N \arrow[l, dashed, bend left, "\exists!\, s"] \\
& V \times W \arrow[ul, "t"] \arrow[u, "b"']
\end{tikzcd}$$

$v \otimes w = t(v,w)$

u is an isomorphism

$t(v,w) = (v,w) + N$

(v,w)

- Question ② What about u?

Known u is surjective

To show $\ker u = 0$

Define s $F(V \times W)/N \to V \otimes W$ $\quad \forall \tau = \Sigma v \otimes w$

$\Sigma x_i (v_i, w_i) + N$ $\qquad v \otimes w \in \operatorname{Im} t$

$(v,w) + N \longmapsto s((v,w) + N)$

First, s is surjective

Next, su and us are surjective

Final. claim $\ker s = 0$

No.

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suppose $s((v,w)+N)=0$ (Inf dim is difficult)

$v\otimes w = t(v,w) \Rightarrow (v,w)\in N$?

<u>Thm</u> let $V, W/\mathbb{F}$ be f.d. Then $V\otimes W \cong F(V\times W)/N \cong \mathrm{span}\{(e_i,f_j)\}$

where $\{e_i\}_{i=1}^{m}$ resp $\{f_j\}_{j=1}^{n}$ is a basis of V resp W

ad hoc, $\dim V\otimes W = \dim V \times \dim W$

☆ Computations on $V\otimes W$

1. $(v+v')\otimes w = t(v+v',w) = t(v,w)+t(v',w) = v\otimes w + v'\otimes w$

2. $(\lambda v)\otimes w = t(\lambda v, w) = \lambda t(v,w) = t(v, \lambda w)$

3. Let $\mathbb{C}/\mathbb{R}$ $\quad \mathbb{C}\otimes_{\mathbb{R}}\mathbb{R}\cong\mathbb{C}$

$V/\mathbb{R}$ $\quad \mathbb{C}\otimes_{\mathbb{R}} V/\mathbb{R}$ $\quad \dim_{\mathbb{R}} \mathbb{C}\otimes_{\mathbb{R}} V = 2\dim_{\mathbb{R}} V$

But $\mathbb{C}\otimes_{\mathbb{R}} V$ has a natural $\mathbb{C}$ space

$c_1\otimes v_1 + c_2\otimes v_2$ (자명) ; $z(c\otimes v) = (zc)\otimes v$